

EECE 2560: Fundamentals of Engineering Algorithms

Department of Electrical and Computer Engineering

Project Sudoku Puzzle

Part b

Complete your program that solves Sudoku puzzles. Your algorithm should be based on recursion and backtracking. Add functions to the board class that:

1. find the next blank cell to fill in a digit,
2. check if a value is legal for a given blank cell, and
3. solve the puzzle using recursion and backtracking

Your algorithm should read each Sudoku board from the file one-by-one, solve it, print the solution, and print the number of recursive iterations needed to solve that board. After all boards have been solved, print the average number of recursive calls needed to solve all the boards.

Your algorithm should minimize the average number of recursive calls needed to solve all the boards. Print the total number of your recursive calls.

(Bonus #1 - 5 points): the top one team that obtains the minimum average number of recursive calls will receive the bonus points.

(Bonus #2 - 5 points): your algorithm only allows to find one solution (Not using exit).